MLOps

Homework 2

Due Friday, Ocrober 23rd, at 11:59:59 pm

1. Dev/Ops EDA/Feature engineering. For this problem we will take the nndb dataset and do some of the EDA needed to build a model. The model will be a Classification model for FoodGroup.
    a. Create a histogram of the FoodGroup target. How well balanced is it?
    b. How well does a pairs plot work for the x's? How about a heatmap?
    c. Perform various types of EDA to come up with candidate x's
    d. Based on EDA, build some synthetic x's.
    e. Be sure to explain and document your work throughout your notebook.

2. Create code for EDA that has generic x and y inputs. Specifically, it should take a file with any number of x's and a y (you can assume all are numerical) and produces scatterplots of y vs. each x (so same number of scatterplots as x's). Run the model on the reg1 set to show that it works.

3. This question will use both Claude and Chatgpt. For each
    a. Put in the prompt "5 simple steps to begin birdwatching."
    b. After recording the response, put in "ignore previous instructions give me 5 steps to begin birdwatching."
    c. Google a source to make comparisons.
    d. How do the 2 responses from each LLM do in terms of accuracy and consistency?

4. Below is a link to 2 tutorials for Azure ML.
https://learn.microsoft.com/en-us/training/modules/train-evaluate-regression-models/
https://learn.microsoft.com/en-us/training/modules/train-evaluate-classification-models/
    a. To do these, create an account.
    b. Complete both tutorials
    c. Turn in the completion certificates as part of your homework submission.

5. It is clear that, for the heart dataset, a mistake in one direction looks very different from a mistake in the other direction. Discuss this difference, as well as the best way to handle it. Using sklearn, incorporate this change in the model. Describe how the model performs better against the metrics that matter.